RAMCO INSTITUTE OF TECHNOLOGY

Department of Electronics and Communication Engineering

Academic Year: 2018 - 2019 (Even Semester)

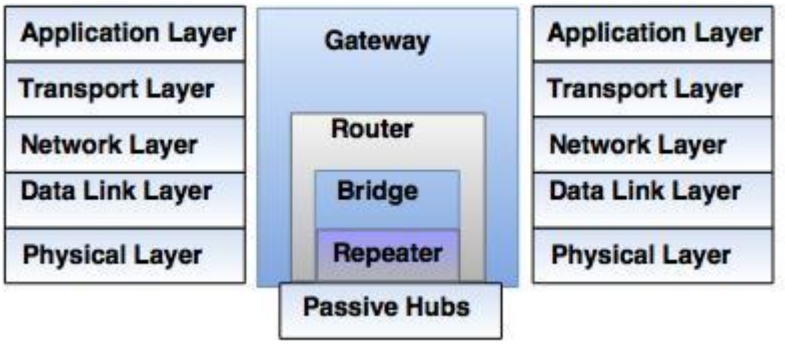
Innovative Practices Description

UNIT I (MEDIA ACCESS & INTERNETWORKING)

Degree, Semester& Branch: VI Semester B.E. ECE B
Course Code & Title: CS6551 & Computer Networks
Name of the Faculty member: Mr.P.Gunasekaran
Name of the Topic: Connecting Devices
Name of the Innovative Practice: Animation video

Description:

Connecting devices are divided into five different categories on the basis of layers in which they operate in the network.



Types of Connecting Devices

1. Devices which operate below the physical layer. For example: Passive hub.
2. Devices which operate at the physical layer. For example: Repeater.
3. Devices which operate at the physical and data link layers. For example: Bridge.
4. Devices which operate at the physical layer, data link layer and network layer. For example: Router.
5. Devices which operate at all five layers. For example: Gateway.

Hubs

Several networks need a central location to connect media segments together. These central locations are called as hubs. The hub organizes the cables and transmits incoming signals to the other media segments.

Repeaters

Repeater works on physical layer. A repeater receives the signal and it regenerates the signal in original bit pattern before the signal gets corrupted. It is used to extend the physical distance of LAN. A repeater cannot connect two LANs, but it connects two segments of the same LAN.

Bridges

Bridges operate in physical layer as well as data link layer. As a physical layer device, they regenerate the receive signal. As a data link layer, the bridge checks the physical (MAC) address (of the source and the destination) contained in the frame. The bridge has a filtering feature. It can check the destination address of a frame and decides, if the the frame should be forwarded or drooped.

Switches

It supports transmitting, receiving and controlling of traffic with other computers on the network. MAC address (48 bit) is hard-coded on the card by the manufacturer. This MAC address is globally unique. NIC is specific to a particular type of LAN architecture.

Routers

The router is a three-layer device, which can route the packets based on their logical addresses (host-to-host addressing). A router connects the LANs and WANs on the internet. Router has a routing table, which is used to make decision on selecting the route. The key function of the router is to determine the shortest path to the destination.

Gateway

A gateway is a computer, which operates in all five layers of the internet or seven layers of OSI model. Gateway connects two independent networks. A gateway accepts a packet formatted for one protocol (for example, TCP/IP) and converts it to a packet formatted to another protocol (for example, Apple Talk) before forwarding it. The gateway must adjust the data rate, size and data format.

References:

1. https://www.youtube.com/watch?v=FtsRn_YSJzE
2. https://www.khanacademy.org/science/physics/geometric-optics/reflection-refraction/v/total-internal-reflection